

Viscosity of Liquids

Pre-Lab Plan
Unit Operations Lab #2
9/17/2019

Group #1, Tuesday, PM section

Team member Names

Marium Murtaza

Adisa Hadzic

Cheryl Slykas

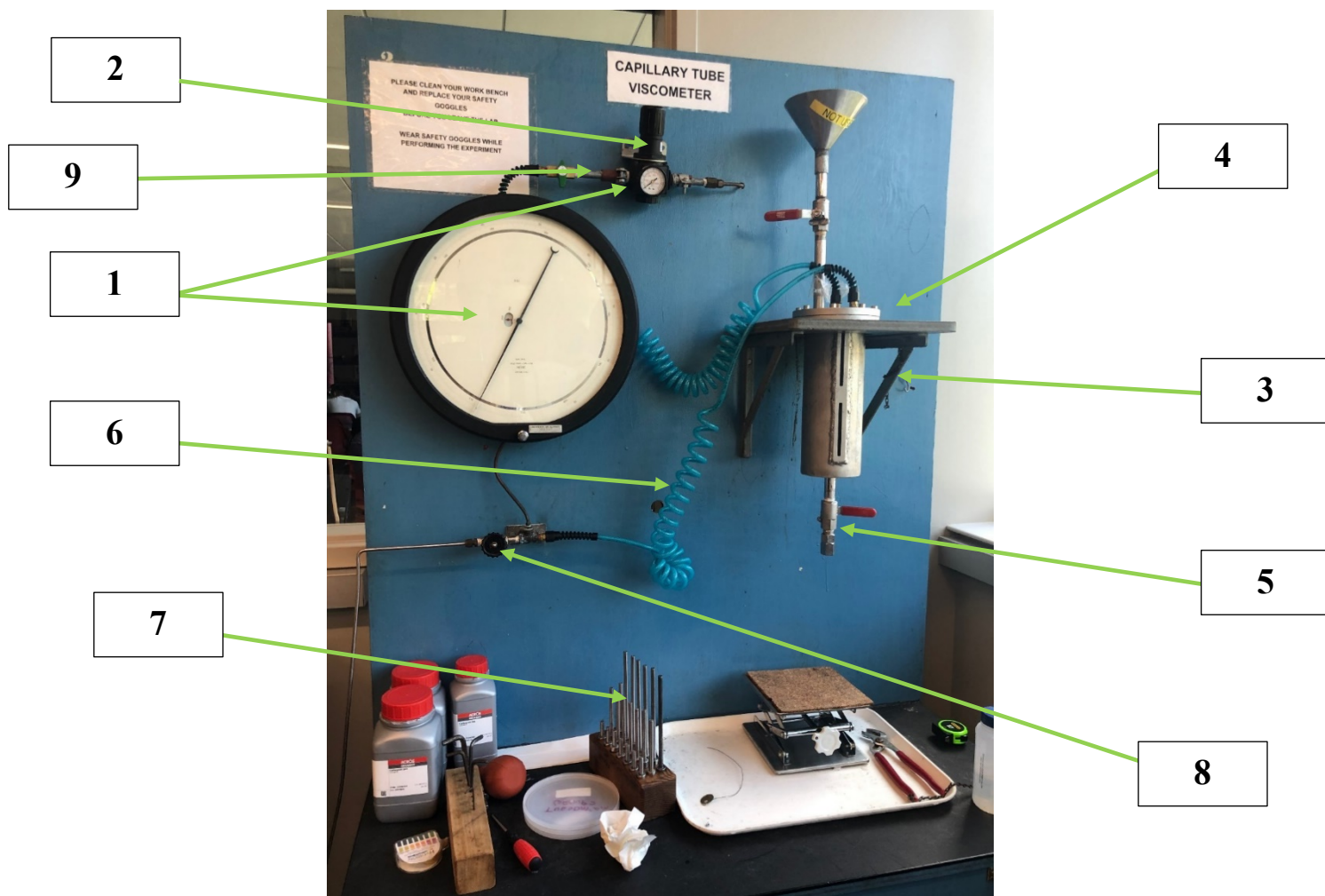
Kellen Krupa

1. Experimental Objective

The objective of this lab is to calculate the shear viscosity in a tube for a non-Newtonian fluid for different tube sizes at different pressures.

2. Experimental

2.1 Apparatus



Main components of the instrument and their functions are described in the table below.

Component	Function
1 - Pressure gauges (2)	One is located above pressure controller to get pressure and the other is bigger that shows is very precise to show how much pressure is present in the vessel.
2 - Pressure controller	When turned will increase or decrease pressure to be entered in vessel.

3 –Pressure Vessel	Filled up with the fluid for which the viscosity is to be measured and it is pressurized with air.
4 – Top Container of Pressure Vessel	Where the fluid is added to get to the vessel- needs to be opened with wrench and closed tight.
5 – Drain Nozzle	When opened, it will release the fluid from the vessel.
6 – Pressure tubing	Blue tube that is attached to main precision gauge and vessel to aid in determining the pressure.
7 - Capillary tubes	Removable tubes of different length to diameter ratios that get attached to the vessel.
8 - Vent valve	It is opened to depressurize the vessel, closed when pressurizing the vessel.
9- Inlet Valve	Allows the pressure to be entered in the more precise and bigger pressure gauge to be read.

2.2 Materials and Supplies

Material	Description
Non-Newtonian fluid	Consists of CARBOPOL, tap water, & NaOH
Stop watch	Times pressure runs
Lab coats	Protect clothing
Safety glasses	Protect eyes
Gloves	For mixing fluid
Scale	Measure mass of fluid
Microsoft Excel	Data collection and tabulation
Power mixer	Mix ingredients for fluid
Graduated cylinder	Used to contain the mixture and to measure for volume measurement of the fluid.
Stainless steel bucket	Contain ingredients to make mixture.
Allen wrench	Use to open screws on pressure vessel
Wrench	Use to change out capillary tubes
Litmus paper	Finds the pH of the mixture to make sure it is between a pH of 6.5-7.5

2.3 Experimental Plan

The Capillary Tube Viscometer

This setup includes small capillary tube that is attached to a small pressurized vessel.

There are different capillary tubes with length/diameter, and they can be removed.

The objective of this experiment is to measure the viscosity of the fluid, that is placed inside the vessel. During the procedure these steps are followed:

- The vessel must be pressurized with air. Apply caution when pressurizing and depressurizing the vessel, and open the valve carefully!
- First, open the vent valve with caution.
- When depressurizing the vessel, it is important to open the vent valve first.
- The vent valve needs to remain while reservoir is filled with the fluid.
- After the vessel is pressurized the vent valve is closed.
- The pressure regulator on the air line is used to apply the constant pressure on the vessel.
- For different trials use different capillary size.
- Flow rate at 4 or 5 different pressure values is measured for each run.
- Pressure must not be greater than 60 psig.
- This experiment is operated at room temperature.
- To prepare CARBPOOL mixer needs to be used, and once solution thickens it is mixed by hand. CARBPOOL also needs to be neutralized with NaOH for the pH of 7 – 7.5 that is tested with the litmus paper.
- For every capillary/new pressure trial, stopwatch needs to be used to record the time, as well as the mass of the solution for calculations of the flow rate.

The non-Newtonian fluid is prepared with 0.8 wt% CARBOPOL solution and water (warm, tap) using power mixer.

The independent variables: pressure and different size capillary tubes.

The dependent variable: viscosity of the CARBOPOL solution.

For the errors calculations we will use trends of the viscosity changes with different capillary size and pressures. They will require 3 trials of different length/diameter of capillary tubes in different pressure conditions. To calculate the error the method of standard deviation is used.

Project Safety Evaluation

Project Title: Viscosity of Liquids

Date: 9/17/19

	<i>Yes/No</i>	
Potential electrical hazards?	yes	If yes, identify conditions: Mixer uses electricity. If water comes in contact with the electrical outlet it can cause fire or electrocution. Shut off/unplug mixer if this happens
Potential mechanical hazards?	yes	If yes, identify conditions: The power mixer mixes the fluids. It can be quickly sped up to a faster rate which can potentially hurt your hands if too close and chemicals may splash.
Potential pressure hazards?	yes	If yes, identify conditions: Pressures not exceeding 60 psig will be used to pressurize the vessel. High pressures in the vessel can cause the tube to burst.
Potential temperature hazards?	no	If yes, identify conditions:
Hazardous/toxic chemicals involved?	yes	If yes, hazards involved: NaOH is a strong base used to neutralize the Carbopol and can be corrosive, cause skin ulcers and burns. Carbopol can be skin and eye irritant.
Gloves required?	yes	If yes, specify type: standard latex
MSDS reviewed by all team members?	yes	List MSDS sheets reviewed: Carbopol 940, NaOH, water
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	20 °C	Room temperature, no hazards
Pressure (psig)	60 psig	The vessel will be pressurized from 0 to 60 psig. If vessel is not depressurized before opening, it can burst.
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
	yes	Depressurize the vessel before changing capillary tubes to avoid accidents. Wear safety goggles and gloves. Use adequate ventilation when using Carbopol and NaOH. Clean up any spills in suitable container. Do not let Carbopol into the environment.

I have read relevant background material for the Unit Operations Laboratory entitled: “Viscosity of Liquids” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Cheryl Slykas 9/17/19
 signature of team member date

Adisa Hadzic 9/17/19
 signature of team member date

Kellen Krupa 9/17/19
 signature of team member date

Marium Murtaza 9/17/19
 signature of team member date